

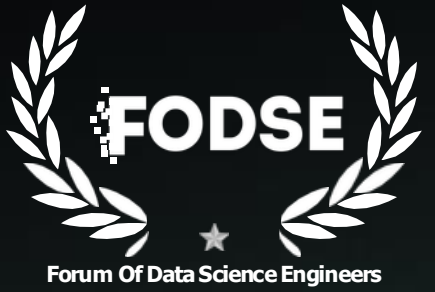


TEAM NAME: Devil Genius

COLLEGE NAME: Sri Venkateswara College Of Engineering

PROBLEM STATEMENT: Virtual Interview Agent

PROBLEM STATEMENT



- **Static Interview Preparation**

Most interview preparation methods rely on reading materials or mock questions without real-time interaction.

- **Lack of Realistic Simulation**

Candidates do not experience dynamic, conversational interview environments similar to real recruiters.

- **No Instant Feedback**

There is limited access to immediate, structured evaluation of spoken responses and communication skills.

- **Absence of Personalization**

Existing platforms rarely adapt questions based on candidate resume, skills, or performance level.

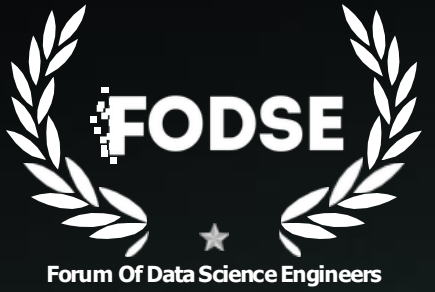
- **Limited Voice-Based Interaction**

Most tools focus on text-based practice and fail to simulate real voice-driven interview scenarios.

- **No Adaptive Questioning**

Systems do not adjust difficulty or generate intelligent follow-up questions based on responses.

SOLUTION



- **AI-Powered Virtual Interview Agent**

A conversational AI agent that autonomously conducts role-based interviews using intelligent dialogue flow.

- **Real-Time Voice Interaction**

Enables candidates to answer via speech, simulating an actual interview environment with live interaction.

- **Resume-Aware Personalization**

Extracts skills and context from uploaded resumes to generate tailored interview questions.

- **Adaptive Questioning Engine**

Dynamically adjusts question difficulty and generates follow-up questions based on candidate responses.

- **Automated Evaluation & Scoring**

Uses AI-driven analysis to assess relevance, clarity, and technical depth, providing structured feedback.

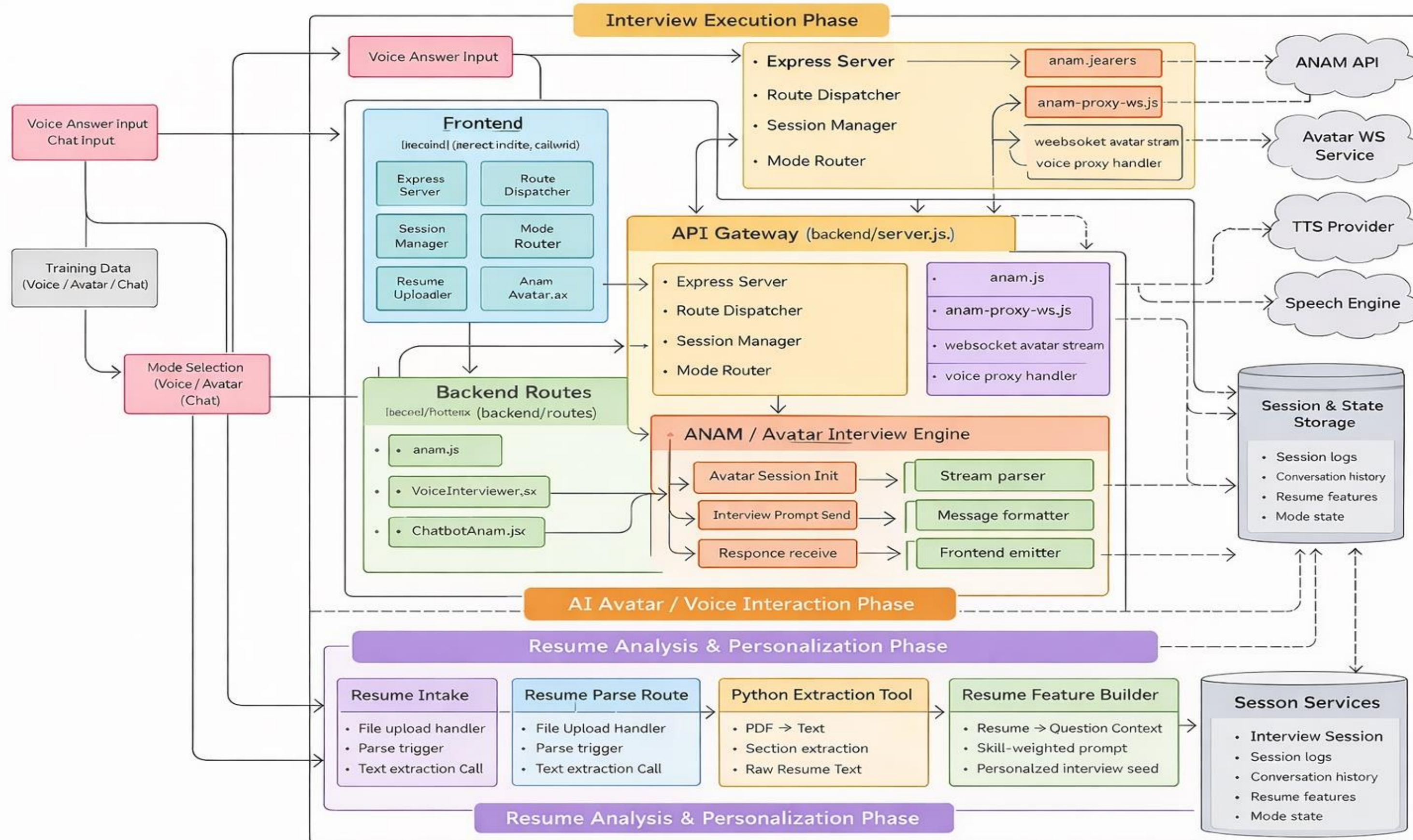
- **Multi-Modal Interview Modes**

Supports voice, chat, and avatar-based interaction for immersive and flexible interview practice.

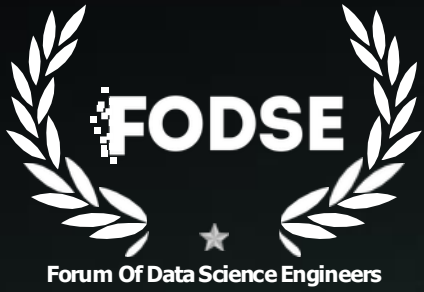


FLOW DIAGRAM

AI VIRTUAL INTERVIEW ARCHITECTURE



TECHNICAL REQUIREMENTS



- **Frontend Requirements**

React (Vite) for responsive UI development

Tailwind CSS for modern styling

WebSocket integration for real-time avatar streaming

Microphone access for voice interaction

- **AI & Integration Requirements**

ANAM API for avatar-based interview interaction

Large Language Model (LLM) for question generation

Speech-to-Text engine for voice input processing

Text-to-Speech (TTS) provider for AI responses

- **Backend Requirements**

Node.js with Express framework

REST API routing and session management

WebSocket proxy for avatar communication

Environment configuration using .env

- **Resume Processing Requirements**

Python-based PDF extraction module

Resume parsing and section extraction

Skill feature builder for personalized question context

- **System Requirements**

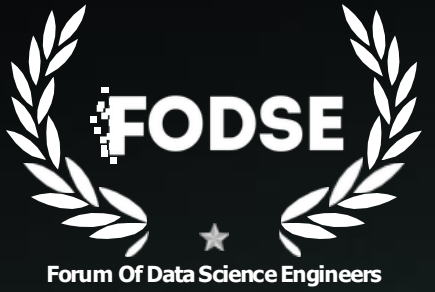
Stable internet connection

Modern browser (Chrome/Edge recommended)

Minimum 8GB RAM for smooth local execution

Secure session & state storage

NOVELTY / USE CASE



- **Agentic Interview Orchestration**

Unlike static mock platforms, the system autonomously manages interview flow using a real-time AI agent.

- **Resume-Aware Question Personalization**

Extracts candidate skills from uploaded resumes to dynamically tailor interview questions.

- **Real-Time Avatar-Based Interaction**

Integrates ANAM avatar with WebSocket streaming for immersive, human-like interview simulation.

- **Multi-Modal Interview Modes**

Supports voice, chat, and avatar-based interaction within a unified framework.

- **Adaptive Questioning Mechanism**

Generates intelligent follow-up questions based on candidate responses and context memory.

- **Student Interview Preparation**

Provides realistic mock interviews for campus placements and job preparation.

- **Corporate Pre-Screening**

Acts as an automated first-level screening tool for recruiters.

- **EdTech Platforms**

Can be integrated into learning platforms for skill assessment and practice.

- **Skill Certification Programs**

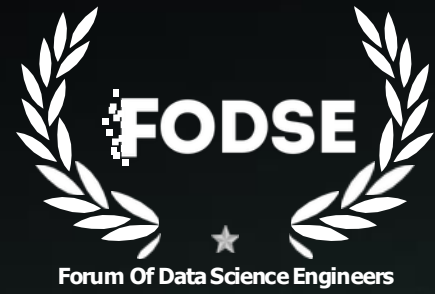
Enables AI-driven assessment for technical skill validation.

- **Career Counseling & Training**

Offers personalized interview practice based on resume and domain expertise.



BUSINESS MODEL / TARGET MARKET



Business Model

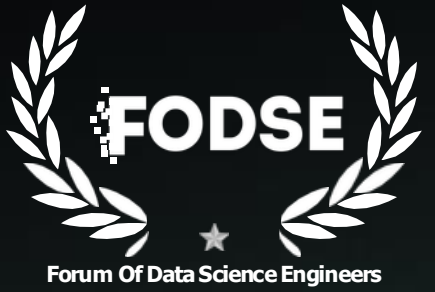
The platform follows a **freemium + subscription model**, offering basic mock interviews for free and advanced features such as resume-based personalization, avatar interviews, detailed analytics, and corporate screening under premium plans.

Additionally, a **B2B licensing model** can be offered to companies, universities, and EdTech platforms for automated candidate screening and skill assessment.

Target Market

- College students preparing for placements
- Job seekers across technical and non-technical domains
- Universities and training institutions
- EdTech platforms
- HR departments and recruitment agencies

TEAM DETAILS



Team member Name	Phone number	College name	Role	Department & Year
Yuvanraaj.C	6379788660	SVCE	Team Leader	CSE-IV
Harisha.S	9445827966	SVCE	Member	IT-III
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Akshaya k	9342180872	SVCE	Member	IT-III
Krithika S	6382641478	SVCE	Member	IT-III
Mohan Raj	9514210203	SVCE	Member	CSE-IV